

Module - 5

1. What is an Expert system? Explain architecture of an expert system with a diagram.
2. Perform the operations i) Algebraic Sum ii) Bounded sum iii) Bounded difference
3. How fuzzy sets are different from crisp sets? Explain following fuzzy operations with examples. i) Fuzzy Union ii) Fuzzy Intersection iii) Fuzzy Complement
4. Mention characteristic features of an expert system and explain its architecture.
5. With appropriate examples, perform any three operations on Fuzzy sets?
6. What is the importance of an Expert System? Mention its characteristic features. Explain Architecture of an expert system.
7. Define the set-theoretic operations for fuzzy sets with suitable examples.
8. Illustrate the architecture of expert system and mention its features.
9. Given the fuzzy sets
 $A = \{0.3/2, 0.4/3, 0.1/4, 0.8/5, 1.0/6\}$
 $B = \{0.7/4, 0.5/5, 1.0/6, 0.02/7, 0.75/8\}$ find $A \cup B$ and $A \cap B$.
10. Briefly explain about typical expert systems.

Module - 4

1. Explain the algorithm to convert WFF to clause with an example
2. Explain Neural net and Genetic learning methods in AI
3. Explain resolution-refutation method in FOPL with an example.
4. Explain
 - a) Goal stack planning.
 - b) Hierarchical planning.
5. What is Goal Stack planning? Solve the following Blocks World problem using goal stack planning algorithm.
6. Consider the following facts and represent it in FOPL:
Ram is a good student. All good students have high grades. All good students with high grades are bright. Show that Ram is bright using Resolution.
7. John likes all kinds of fruits. Orange is a fruit. Prove that John likes Orange in two ways.
8. Explain various learning techniques in AI? What methods are used for learning?
9. Explain the various forms of learning.
10. What is goal stack planning algorithm? Explain with an example.

Module - 3

1. Express the following concepts as semantic network structure with interconnected nodes and labeled arcs: Company Hamen is a software development company. Three departments within the company are Sales, Administration and Programming. Ansu is the manager of Programming. Deepthi and Vini are Programmers. Deepthi is married to Vipin. Vipin is an editor for journal of Computing. They have three children and live in Elm street. Deepthi wears glasses and is five feet four inches tall.
2. What is Conceptual dependency theory? What are its components?
3. Explain MINMAX algorithm with the help of an example.

4. What is Conceptual dependency Theory? What are its components?
5. Express the following concepts as a semantic network structure with interconnected nodes and labeled arcs:
IGNOU is an open university. It is located in Delhi. IGNOU has a Vice chancellor and has regional centers. IGNOU has a school of studies which offers Academic Programmes such as MCA and BCA. MCA has Syllabus and CS-51 is a MCA syllabus. School of studies has a Director.
6. Discuss Alpha-Beta cut-off algorithm with the help of an example.
7. Explain alpha-beta pruning algorithm with an example.
8. Explain about Frames and Conceptual Dependency.
9. Explain alpha-beta pruning and determine which of the branches in the game tree below will be pruned if we apply alpha-beta pruning to solve the game (Assume that the maximising player plays first).
10. a) List advantages and disadvantages of semantic networks.
b) Describe the different types of semantic networks with examples.